PAVITHRA RADHAKRISHNAN
Senior Embedded Software Engineer | AUTOSAR BSW | COM | Diagnostics
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Portfolio: View my portfolio

PROFESSIONAL SUMMARY

Results-driven Senior Embedded Software Engineer with **4.5 years of experience** in automotive embedded software development at **Bosch Global Software Technologies**, contributing to **Jaguar Land Rover** steering programs. Hands-on expertise in AUTOSAR COM, BSW configuration, RTE integration, DCM/DEM, Embedded C, UDS diagnostics, and CAN/CAN FD/FlexRay communication. Experienced across the automotive V-cycle, including requirements analysis, implementation, unit and integration testing, debugging, root-cause analysis, static code analysis, and release support. Demonstrated component ownership, test automation, requirements traceability, and technical mentoring.

TECHNICAL SKILLS

- **AUTOSAR:** COM, DCM, DEM, BSW Configuration, RTE Development
- **Programming:** Embedded C, C, Python, CAPL
- **Automotive Communication:** CAN, CAN FD, FlexRay
- **Diagnostics:** UDS (ISO 14229), DID Development, Diagnostic Routines, DTC Analysis
- **Functional Safety:** ISO 26262, E2E Protection for Communication Fault Detection
- **Static Analysis & Coding Standards:** AbsInt Astrée, MISRA C
- **Testing & Validation:** TESSY, CANoe, CANape, Vector vTestStudio, vCAR, Unit Testing, Integration Testing
- **Debugging & Analysis:** TRACE32/Lauterbach, GLIWA T1, Root-Cause Analysis, Runtime Analysis
- **Tools:** AEEE Pro, SystemDesk, IBM DOORS, Enterprise Architect, Git, Bitbucket, JIRA, VS Code
- **Processes:** Automotive V-Cycle, ASPICE, Agile, Requirements Traceability

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EXPERIENCE

Bosch Global Software Technologies | April 2022 – Present

Senior Engineer – Embedded Software Development

Developing automotive steering software for Jaguar Land Rover (JLR), covering requirements analysis, software design, implementation, testing, debugging, maintenance, and release support.

Project: JLR SCU4 D6

Communication protocol: CAN FD

- Implemented communication signals and signal groups from customer DBC specifications, covering AUTOSAR BSW configuration, RTE integration, application-layer implementation, and testing.
- Owned the communication component and achieved 100% requirements traceability across software design, implementation, unit testing, and component testing, supporting ASPICE process compliance.
- Implemented E2E protection for communication signals to support communication-fault detection in line with project functional safety requirements.
- Enhanced and maintained CANoe-based Rest Bus Simulation using CAPL to support communication testing.
- Performed unit testing using TESSY and vCAR, and communication and diagnostic validation using CANoe and CANape.
- Debugged embedded software using TRACE32 and conducted runtime and timing analysis using GLIWA T1.
- Performed static code analysis using AbsInt Astrée and resolved MISRA C compliance findings.
- Analyzed requirements, conducted impact analysis, and maintained traceability using IBM DOORS.
- Introduced project-wide author self-review practices, reducing review and rework time by 30%.
- Conducted code and design reviews and mentored junior engineers.
- Demonstrated the RB Security Access feature (Gen2) in Germany, supporting security validation.

Project: JLR BWI

Communication protocol: FlexRay

- Performed software maintenance, enhancement, defect analysis, and issue resolution.
- Developed customer-specific diagnostic Data Identifiers (DIDs) and diagnostic routines.
- Analyzed software architecture and communication flows.
- Took ownership of MONID configuration within DEM-related activities and delivered project commitments under tight deadlines.

AUTOMATION & INNOVATION

- Developed reusable COM validation automation using Vector vTestStudio and CAPL, reducing validation effort by 90%.
- Developed Python-based part-number automation to minimize manual errors.

CERTIFICATIONS AND PERSONAL DEVELOPMENT

- IREB Certified Professional for Requirements Engineering (CPRE) – Foundation Level
- ISTQB Certified Tester – Foundation Level
- Diagnostics Skill Acceleration Program
- COM and Infrastructure Skill Acceleration Program
- Technical Career Qualification
- Be10X AI Career Accelerator Program

AWARDS & RECOGNITION

- Top Performer – DCM Skill Acceleration Program
- Best Performer – Diagnostics and COM Competency Development Program
- Multiple Bosch Quarterly Awards, Shabash Awards, Rockstar Awards and Team Recognition Awards
- Crowdsourcing Champion Award for completing activities across projects
- Received recognition for building “Career Coach Platform” using AI in the hackathon conducted by Be10X

EDUCATION

- **Bachelor of Engineering – Electronics and Communication Engineering** | August 2017 – May 2021 | **CGPA 9.18/10**
  Kongu Engineering College, Perundurai
- **Higher Secondary Education |** June 2016 – May 2017 | **Percentage 95.67%**
  SVN Matriculation Higher Secondary School, Erode
- **SSLC |** June 2014 – May 2015 | **Percentage 99.6%**
  SVN Matriculation Higher Secondary School, Erode
